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Synthesis and Characterization of a Quantum Magnet with the Spinel Structure

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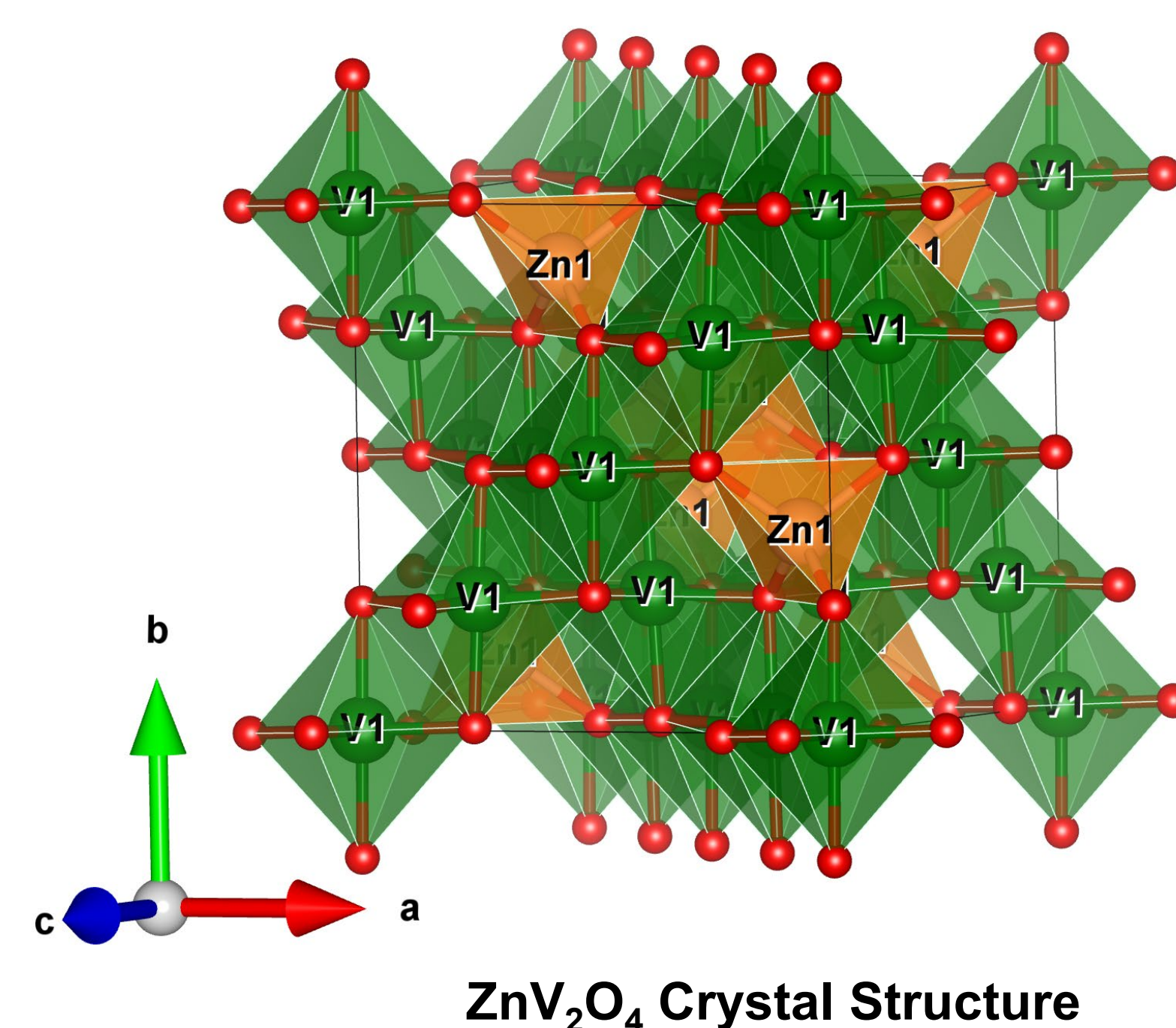
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Introduction: Spinel compounds & their properties

- Spinel compounds are materials composed of an oxide ion and a metal cation with a cubic crystal structure.
- The general formula is AB_2O_4 , where A lies in tetrahedral sites and B lies in octahedral sites.
- $Li_xZn_{(1-x)}V_2O_4$ is the spinel compound of particular interest due to its potential use in battery technology and electrochemical energy storage.
- The aim of this study is to investigate the effects of introducing disorder on the magnetic behavior of the parent ZnV_2O_4 spinel.



Methods

Solid State Synthesis

A. Starting Compounds:



B. Chemical Reaction:



Where x is a value between 0 and 1, representing ratio of Zn replaced by Li

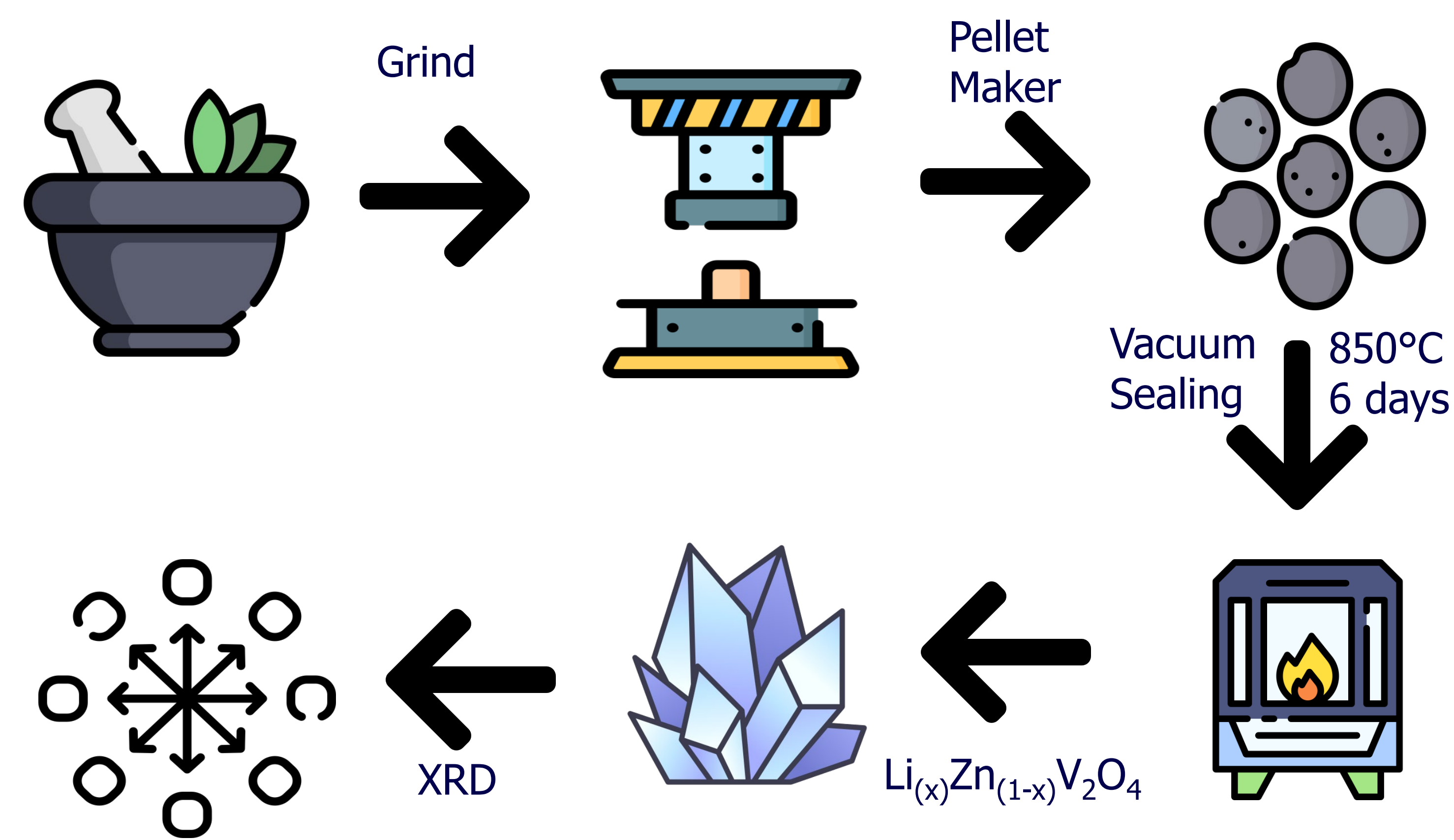
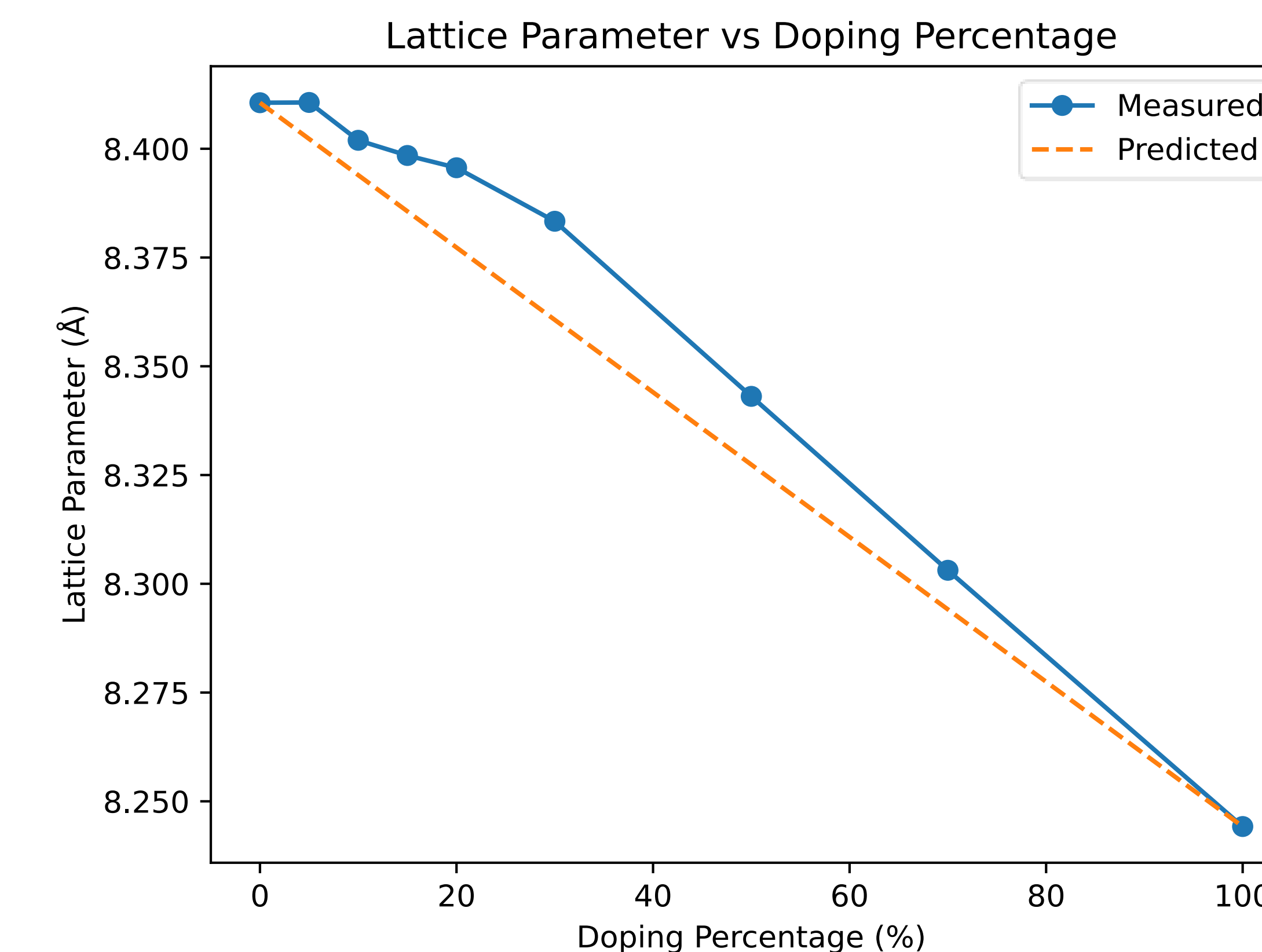
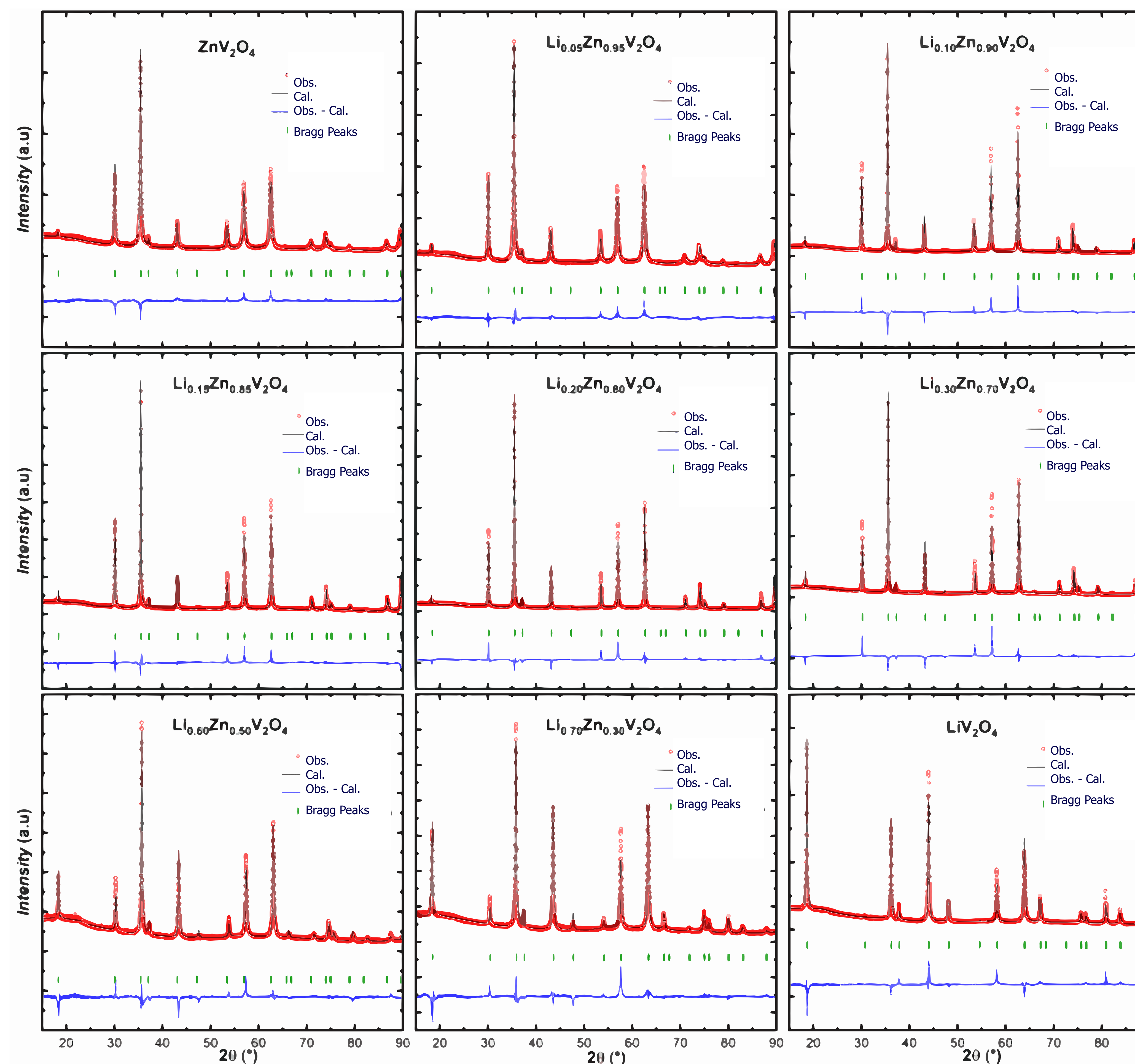


Figure: Schematic Diagram of the Solid-State Reaction & Diffraction Methods

Results, Analysis, and Discussions



Analysis and Discussions

- The structural characterization of $Li_xZn_{(1-x)}V_2O_4$ were carried out using X-ray diffraction analysis.
- A linearly decreasing trend of lattice constant was observed with the increase in Li-doping to ZnV_2O_4 .
- $Li_xZn_{(1-x)}V_2O_4$ appear to follow Vegard's rule with respect to changes in lattice parameter on doping.
- The average lattice deviation was measured to be 0.0094 Å.
- The decreasing trend can be attributed to the smaller atomic radius of Li as compared to Zn.
- The slight deviation suggests that the interaction between the dopants and the host material is not weak.
- The deviation can be attributed to the difference in atomic radii between Li and Zn, leading to lattice strain and distortion.

Conclusions

- Li-doping in ZnV_2O_4 results in a linear decrease in lattice constant, following Vegard's Law.
- A small deviation from Vegard's Law indicates a considerable interaction between dopants and the host material.
- Lattice distortion is likely due to the difference in atomic radii between Li and Zn.
- Further, thermodynamic studies will be carried out to understand the effect of Li doping in the series $Li_xZn_{(1-x)}V_2O_4$.

References

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